# Emilio Rodea

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## EDUCATION

**Bachelor of Science in Mechanical Engineering – GPA 3.51** **Graduation Date: May 2028**
**California State Polytechnic University, Pomona** **Pomona, CA**

La Costa Canyon High School – GPA 4.0 **Graduation Date: June 2023**

## WORK EXPERIENCE

**Mechanical Design Intern** **June 2026 – Aug 2026**
ZMicro San Diego, CA

- **Designed and fabricated a sheet-metal split-shell enclosure** for a ruggedized removable storage module (TP1/TP1E) using DFM/DFA principles, meeting MIL-STD-461G EMI and MIL-STD-810G/DO-160G shock/vibration requirements.
- **Designed and revised SolidWorks parts and assemblies for rugged LCD displays and servers**, including rear bezels, sheet-metal parts, fastener fit-ups, producing GD&T-dimensioned drawings with PDM vault management.
- **Conducted root cause failure and thermal/power analysis** for rugged hardware, created ECOs for engineering change control, and supported ICD documentation and configuration control for defense customer programs specifications.
- **Authored a technical brief proposing AI integration** into ZMicro's Change order's, DCP, DVT, and PDM workflows.

**Project Engineer Intern** **Summer 2025**
Filanc Construction On Site – Laguna Beach, CA

- **Leveraged AutoCAD** to cross-check elevations, layouts, and site conditions against shop drawings, ensuring accuracy across the construction environment and consistency with project and contract specifications.
- **Created project deliverables,** including quantity take-offs, budget estimates, detailed work packages, and quality control procedures, ensuring accuracy and compliance with project requirements and contract specifications.
- **Prepared and processed** submittals, RFIs, and change orders, supporting communication between field teams and subcontractors **while maintaining correct contract bid documentation to ensure cost control and schedule adherence**.
- **Solicited and evaluated subcontractor quotes** by distributing detailed work packages, analyzing initial bid estimates, and finalizing contractors' scopes of work, **saving the company an estimated $50K+ versus initial estimates**.

**Social Media Marketing Intern** **Summer 2024**
**Road Runner Sports** San Diego, CA

- **Assisted in copywriting and social media content creation**, ensuring engaging and brand-consistent messaging.
- **Analyzed and organized advertising data**, providing insights to brands for optimizing ad performance.
- **Improved ad performance and engagement by 20% by leading** initiatives to incorporate **AI-driven marketing strategies**.

## BUSINESS EXPERIENCE

**Amazon Brand Owner - UniRodea** **April 2025 - Present**

- Developed a home-decor product line on Amazon **using AI tools** for concept creation, branding, and market positioning.
- **Managed Amazon PPC campaigns**, optimizing keywords and budgets to increase visibility and sales performance.

**Etsy Shop Owner** **June 2024 - Present**

- **Scaled my Etsy store to $40K+ in revenue** by leveraging SEO, digital marketing, consumer insights, and **utilizing AI.**
- Designed a custom carabiner in SolidWorks and oversaw mold development and manufacturing for product launch.

## PROJECTS

**6-DOF Robotic Arm – LeRobot Imitation Learning Platform** **2026 - Present**

- **Designing and fabricating a 6-DOF robotic arm from scratch in SolidWorks** with servo-driven actuation for teleoperated control, building a Python/PyTorch/LeRobot stack to train an imitation-learning policy from demonstration data.

## TECHNICAL SKILLS

**Computer** | AutoCAD, SolidWorks, MATLAB, Bluebeam, Excel, Python, C/C++, Arduino, GRBL, G-Code
**Skills** | DFM/DFA, Lean Manufacturing, GD&T, Multi-Axis Motion Control, ECO, Sheet Metal Fabrication, Root Cause Analysis, Project Management, Construction Take-offs, Problem Solving & Initiative, Collaboration & Team member